

UEI516: ANALYTICAL INSTRUMENTATION

L T P Cr
3 0 0 3.0

Course objectives: To introduce the concept of analytical Instrumentation, methods, techniques and applications.

Introduction: Introduction to instrumental analysis-classification and its advantages, Sampling systems for gas analysis and liquid analysis.

Spectrometry: Introduction to atomic absorption spectrometer, emission spectrometer UV-visual spectrometer, infrared spectrometer, excitation sources: arc and spark, Nuclear magnetic resonance spectrometer, Mass spectrometry, biomedical applications of spectrometry.

Chromatography: Introduction to Chromatographic techniques, Liquid chromatography, Gas chromatography, Applications of chromatography. Introduction to optical Techniques and their Working, turbidimetry, Nephelometry, Polarimetry, Refractometry.

X-ray Analytical Methods: Introduction to X-ray spectral analysis, Fluorescence X-ray spectrometer Wavelength dispersive devices, Energy dispersive devices, Detectors, Scanning electron microscope, X-ray diffractometer, X ray absorption spectrometer Applications of X ray analytical methods in biomedical, industrial applications.

Potentiometry: Potential and standard potential, ion selective electrode, Glass electrode, Gas sensing electrode. Application of potentiometry.

Course Learning Outcomes (CLO): After the completion of the course student will be able to:

1. Demonstrate the knowledge of concept of spectrometry and optical techniques.
2. Elucidate the working of chromatography, elemental analyzer.
3. Illustrate the working of X- ray diffractometer and scanning electron microscope.
4. Exhibit the concept of potentiometry and its applications.
5. Exhibit the knowledge of working principle and applications of different electrodes.

Text Books:

1. Braun, R.D., *Introduction to Instrumental Analysis*, Mc-Graw Hill (2008).
2. Khandpur, R.S., *Handbook of Biomedical Instrumentation*, Tata McGraw Hill (2000)
3. Mathur, R.P., *Water and Waste Water Testing Laboratory Manual*, Nem Chand and Brothers (1982).
4. Patranabis D. *Principles of Industrial & Instrumentation*, Tata McGraw Hill (1998)

Evaluation Scheme:

S.No.	Evaluation Elements	Weightage (%)
1.	MST+EST	70
2.	Sessionals (May include Assignments/Projects/Tutorials/Quizzes/Lab Evaluations)	30